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AQA A Level Further Mathematics (7367)

Topic **DB: Networks**

Sub topic **DB3: Route Inspection Problems**

Topic Questions

AQA A Level Further Mathematics (7367)

Question Paper

Topic	**DB: Networks**
Sub topic	**DB3: Route Inspection Problems**

To be used by all students and teachers for AQA A Level Further Mathematics (7367) for exams from 2025 onwards.

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Name : ____________________

Date : ____________________

Time : ____________________

Total Marks Available : _____

Total Marks Received : _____

Q1.

A communications company is conducting a feasibility study into the installation of underground television cables between 5 neighbouring districts.

The length of the possible pathways for the television cables between each pair of districts, in miles, is shown in the table.

The pathways all run alongside cycle tracks.

	Billinge	**Garswood**	**Haydock**	**Orrell**	**Up Holland**
Billinge	-	2.5	***	4.3	4.8
Garswood	2.5	-	3.1	***	5.9
Haydock	***	3.1	-	6.7	7.8
Orrell	4.3	***	6.7	-	2.1
Up Holland	4.8	5.9	7.8	2.1	-

(a) Give a possible reason, in context, why some of the table entries are labelled as *****.

(1)

(b) As part of the feasibility study, Sally, an engineer, needs to assess each possible pathway between the districts. To do this, Sally decides to travel along every pathway using a bicycle, starting and finishing in the same district. From past experience, Sally knows that she can travel at an average speed of 12 miles per hour on a bicycle.

Find the minimum time, in minutes, that it will take Sally to cycle along every pathway.

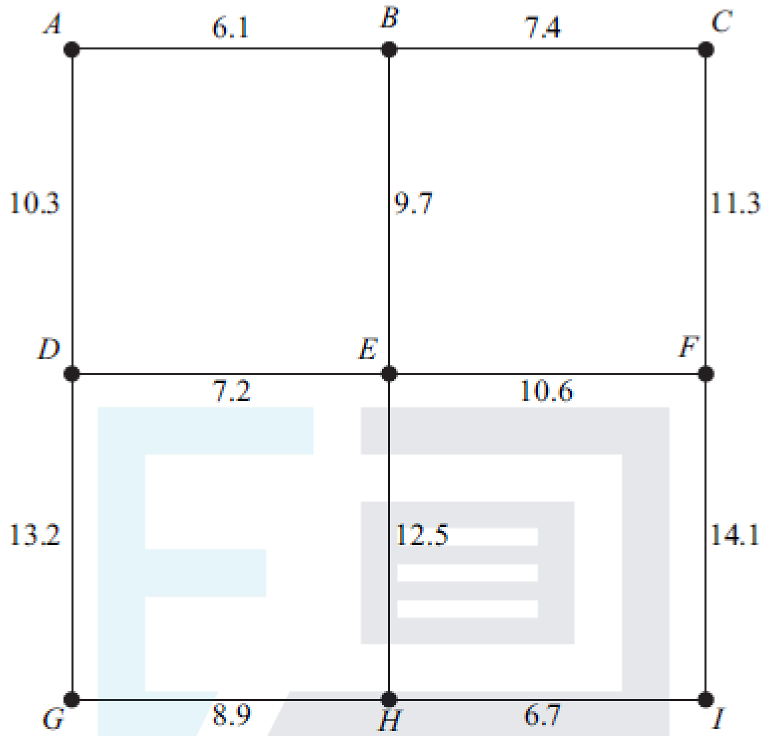
(4)

(Total 5 marks)

Q2.

The following network shows the lengths, in miles, of roads connecting nine villages, A, B, …, I.

A delivery man lives in village A and is to drive along all the roads at least once before returning to A.



Total length of all the roads is 118 miles

(a) Find the length of an optimal Chinese postman route around the nine villages, starting and finishing at A.

(5)

(b) For an optimal Chinese postman route corresponding to your answer in part (a), state:

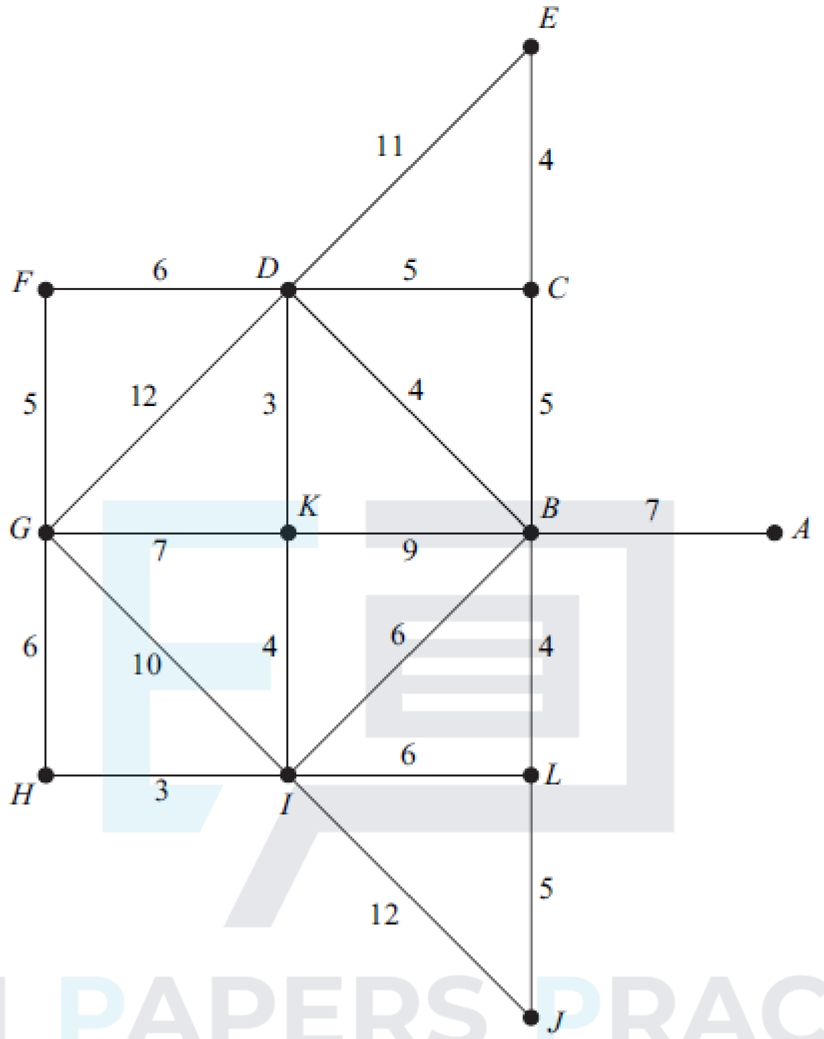
(i) the number of times village E would be visited;

(ii) the number of times village I would be visited.

(2)
(Total 7 marks)

Q3.

The network shows the times, in minutes, taken by police cars to drive along roads connecting 12 places, A, B, …, L.



Total of all times = 134 minutes

On a particular day, there are three police cars in the area at A, E and J. There is an emergency at G and all three police cars drive to G.

Each day, a police car has to drive along each road at least once, starting and finishing at A.

For an optimal Chinese postman route:

(a) find the driving time for the police car;

(6)

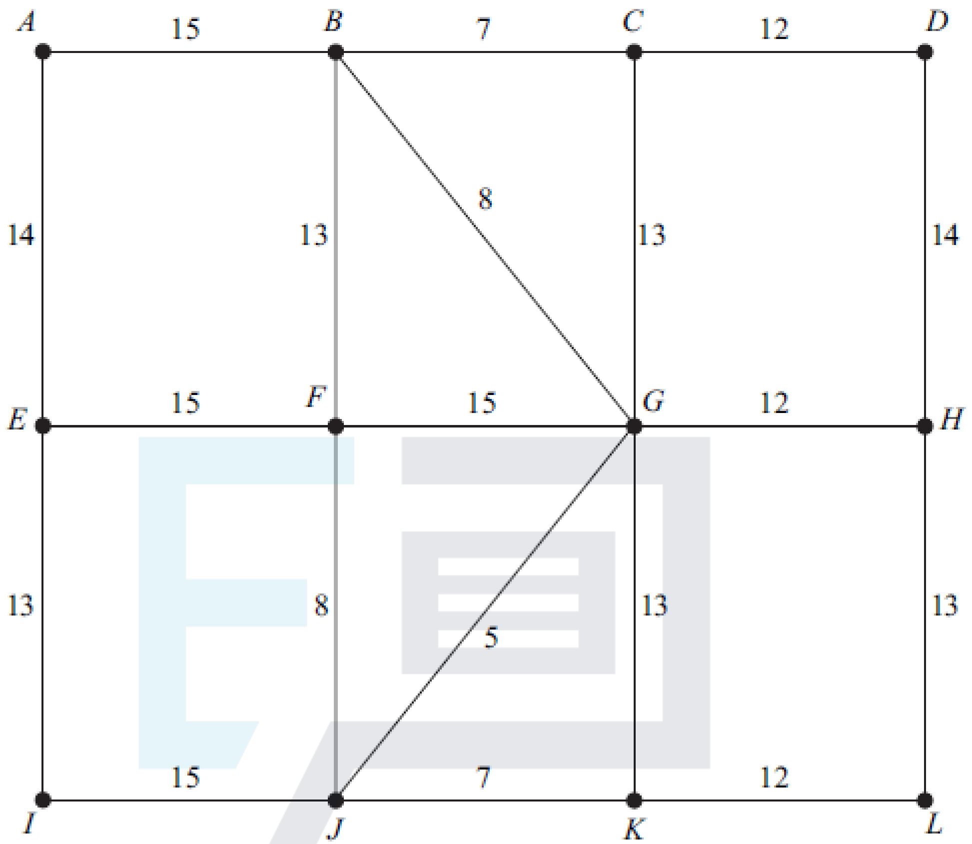
(b) state the number of times that the vertex B would appear.

(1)

(Total 7 marks)

Q4.

The following network shows the times, in minutes, taken by a policeman to walk along roads connecting 12 places, A, B, ..., L, on his beat. Each day, the policeman has to walk along each road at least once, starting and finishing at A.



The total of all the times in the network is 224 minutes.

(a) Find the length of an optimal Chinese postman route for the policeman.

(5)

(b) State the number of times that the vertex J would appear in a route corresponding to the length found in part (a).

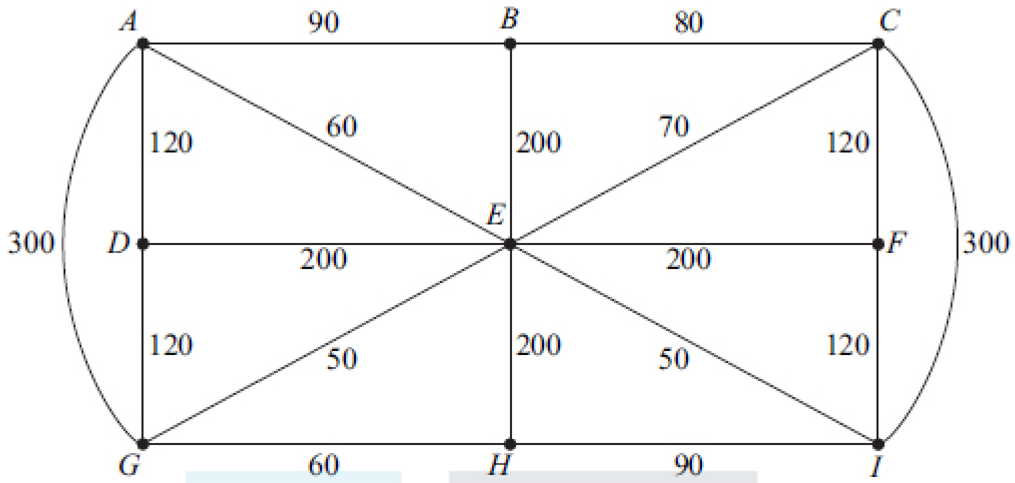
(1)

(Total 6 marks)

Q5.

The network below shows some streets in a town. The number on each edge shows the length of that street, in metres.

Leaflets are to be distributed by a restaurant owner, Tony, from his restaurant located at vertex B. Tony must start from his restaurant, walk along all the streets at least once, before returning to his restaurant.



The total length of the streets is 2430 metres.

(a) Find the length of an optimal Chinese postman route for Tony.

(5)

(b) Colin also wishes to distribute some leaflets. He starts from his house at H, walks along all the streets at least once, before finishing at the restaurant at B.

Colin wishes to walk the minimum distance. Find the length of an optimal route for Colin.

(1)

(c) David also walks along all the streets at least once. He can start at any vertex and finish at any vertex. David also wishes to walk the minimum distance.

(i) Find the length of an optimal route for David.

(1)

(ii) State the vertices from which David could start in order to achieve this optimal route.

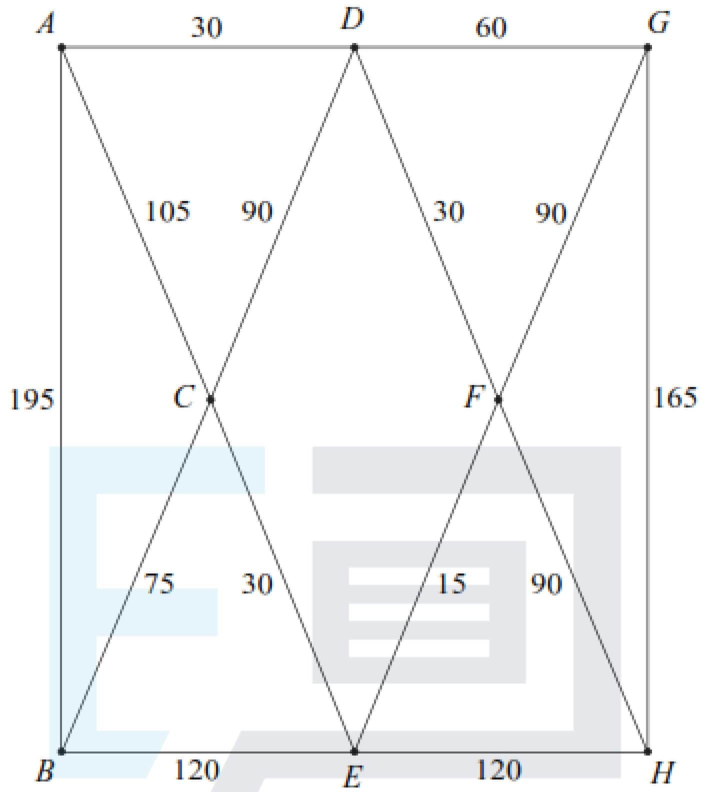
(1)

(Total 8 marks)

Q6.

Norris delivers newspapers to houses on an estate. The network shows the streets on the estate. The number on each edge shows the length of the street, in metres.

Norris starts from the newsagents located at vertex A, and he must walk along all the streets at least once before returning to the newsagents.



The total length of the streets is 1215 metres.

(a) Give a reason why it is not possible to start at A, walk along each street once only, and return to A.

(1)

(b) Find the length of an optimal Chinese postman route around the estate, starting and finishing at A.

(5)

(c) For an optimal Chinese postman route, state:

(i) the number of times that the vertex F would occur;

(1)

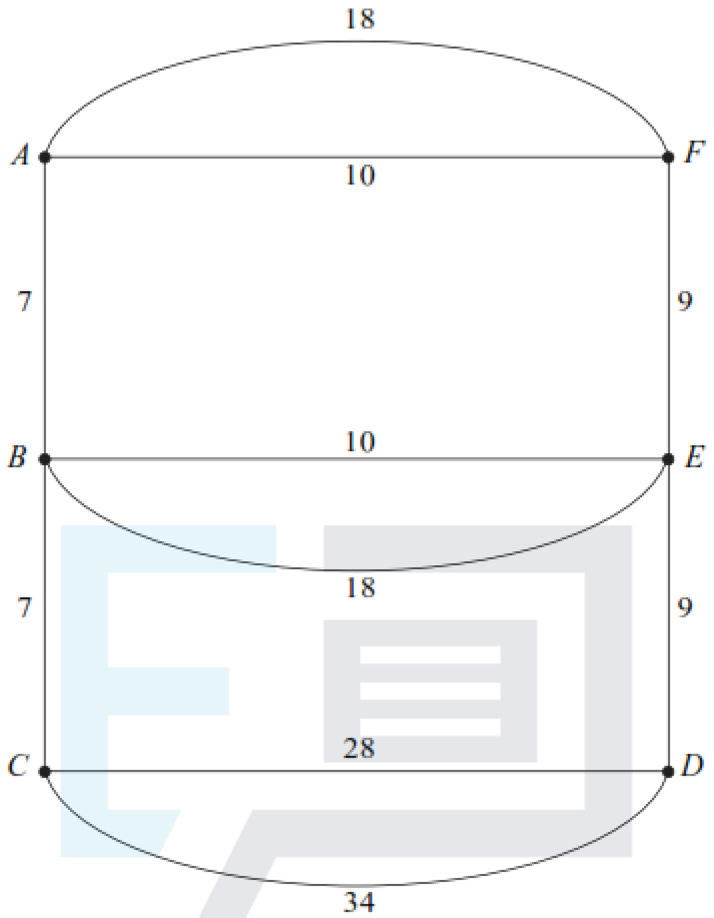
(ii) the number of times that the vertex H would occur.

(1)

(Total 8 marks)

Q7.

A council is responsible for gritting main roads in a district. The network shows the main roads in the district. The number on each edge shows the length of the road, in kilometres. The gritter starts from the depot located at point A, and must drive along all the roads at least once before returning to the depot.



Total = 150 km

(a) Find the length of an optimal Chinese postman route around the main roads in the district, starting and finishing at A.

(5)

(b) Zac, a supervisor, wishes to inspect all the roads. He leaves the depot, located at point A, and drives along all the roads at least once before finishing at his home located at point C. Find the length of an optimal route for Zac.

(2)

(c) Liz, a reporter, intends to drive along all the roads at least once in order to report on driving conditions. She can start her journey at any point and can finish her journey at any point.

(i) Find the length of an optimal route for Liz.

(2)

(ii) State the points from which Liz could start in order to achieve this optimal route.

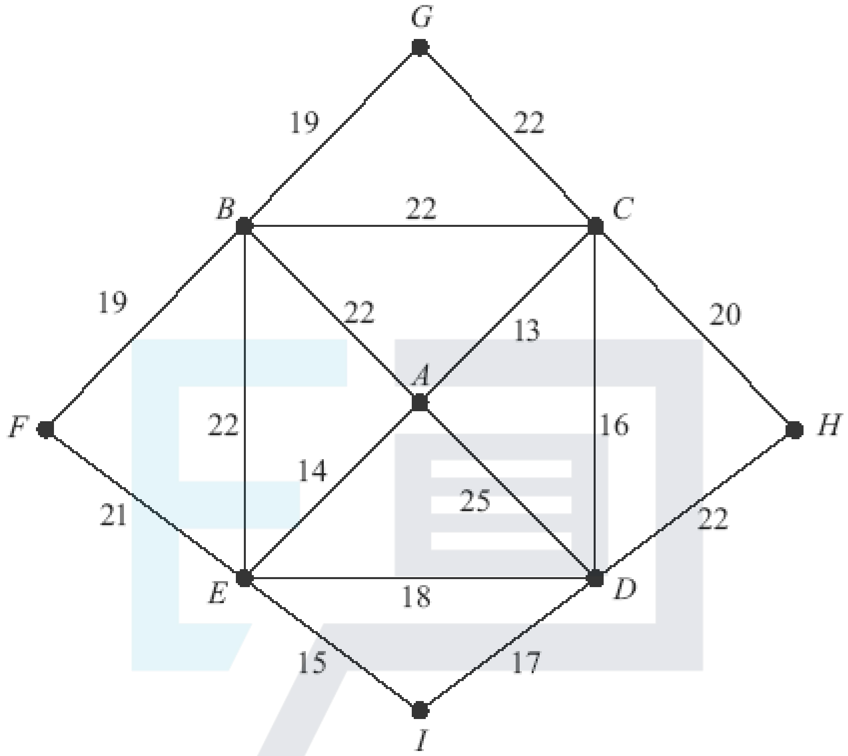
(1)
(Total 10 marks)

Q8.

In Paris, there is a park where there are statues of famous people; there are many visitors each day to this park. Lighting is to be installed at nine places, $A, B, \ldots, I$, in the park.

The places have to be connected either directly or indirectly by cabling, to be laid alongside the paths, as shown in the diagram.

The diagram shows the length of each path, in metres, connecting adjacent places.



Total length of paths = 307 metres

(a) (i) Use Prim's algorithm, starting from A, to find the minimum length of cabling required.

(5)

(ii) State this minimum length.

(1)

(iii) Draw the minimum spanning tree.

(2)

(b) A security guard walks along all the paths before returning to his starting place. Find the length of an optimal Chinese postman route for the guard.

(6)
(Total 14 marks)

Q9.

The network below shows 13 towns. The times, in minutes, to travel between pairs of towns are indicated on the edges.

The total of all the times is 384 minutes.

(a) Find the travelling time of an optimum Chinese postman route around the network, starting and finishing at M.

(6)

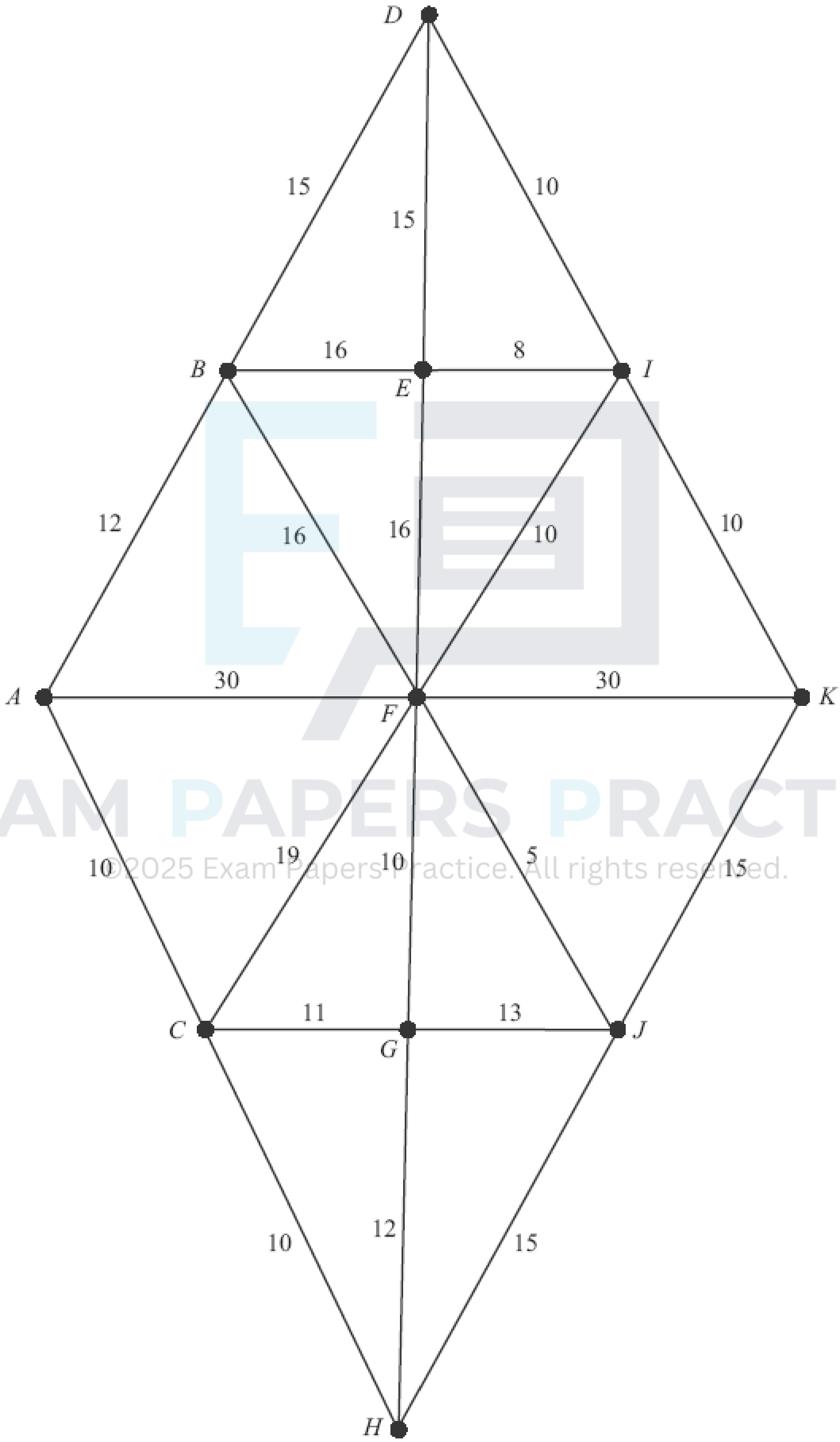
(b) State the number of times that the vertex F would appear in a corresponding route.

(1)

(Total 7 marks)

Q10.

The network shows 11 towns. The times, in minutes, to travel between pairs of towns are indicated on the edges.



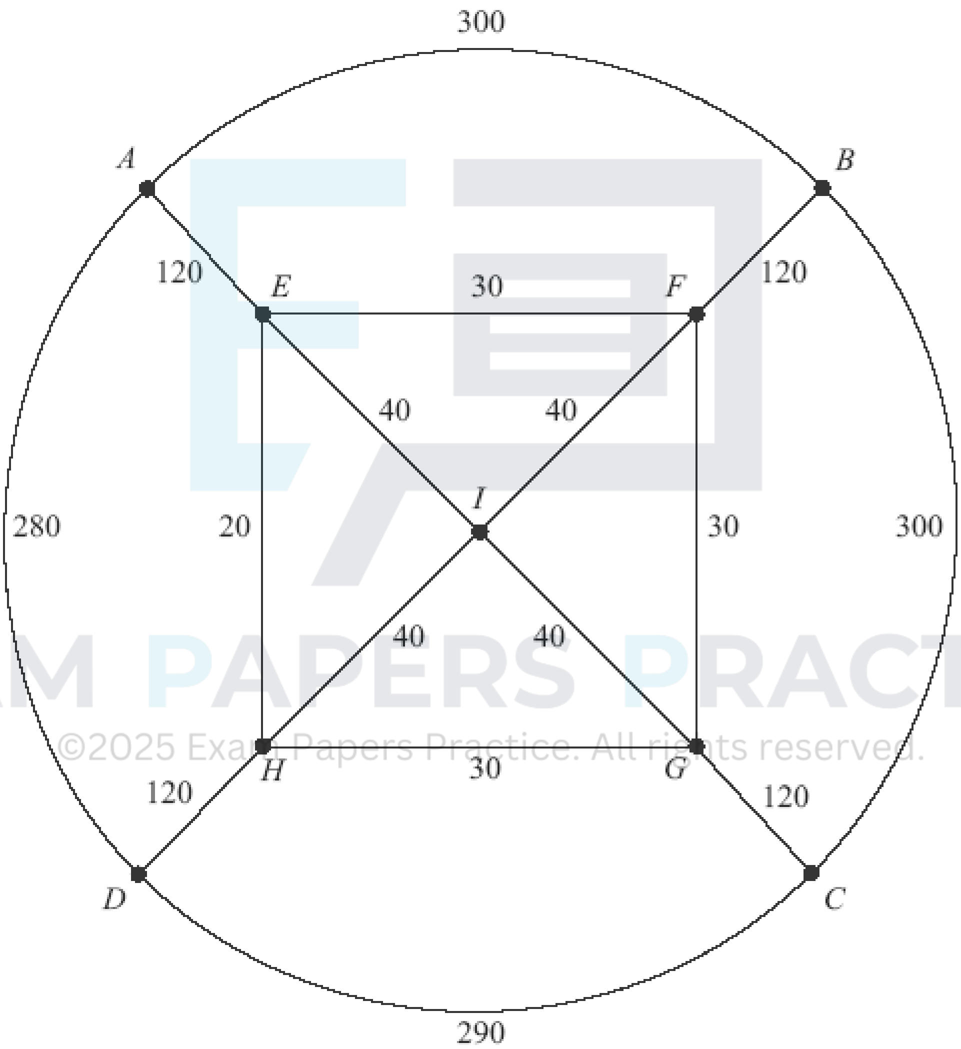
The total of all of the times is 308 minutes.

Find the length of an optimum Chinese postman route around the network, starting and finishing at A. (The minimum time to travel from D to H is 40 minutes.)

(Total 5 marks)

Q11.

The diagram shows a network of sixteen roads on a housing estate. The number on each edge is the length, in metres, of the road. The total length of the sixteen roads is 1920 metres.



Total Length = 1920 metres

(a) Chris, an ice-cream salesman, travels along each road at least once, starting and finishing at the point A. Find the length of an optimal 'Chinese postman' route for Chris.

(6)

(b) Pascal, a paperboy, starts at A and walks along each road at least once before finishing at D. Find the length of an optimal route for Pascal.

(2)

(c) Millie is to walk along all the roads at least once delivering leaflets. She can start her journey at any point and she can finish her journey at any point.

(i) Find the length of an optimal route for Millie.

(2)

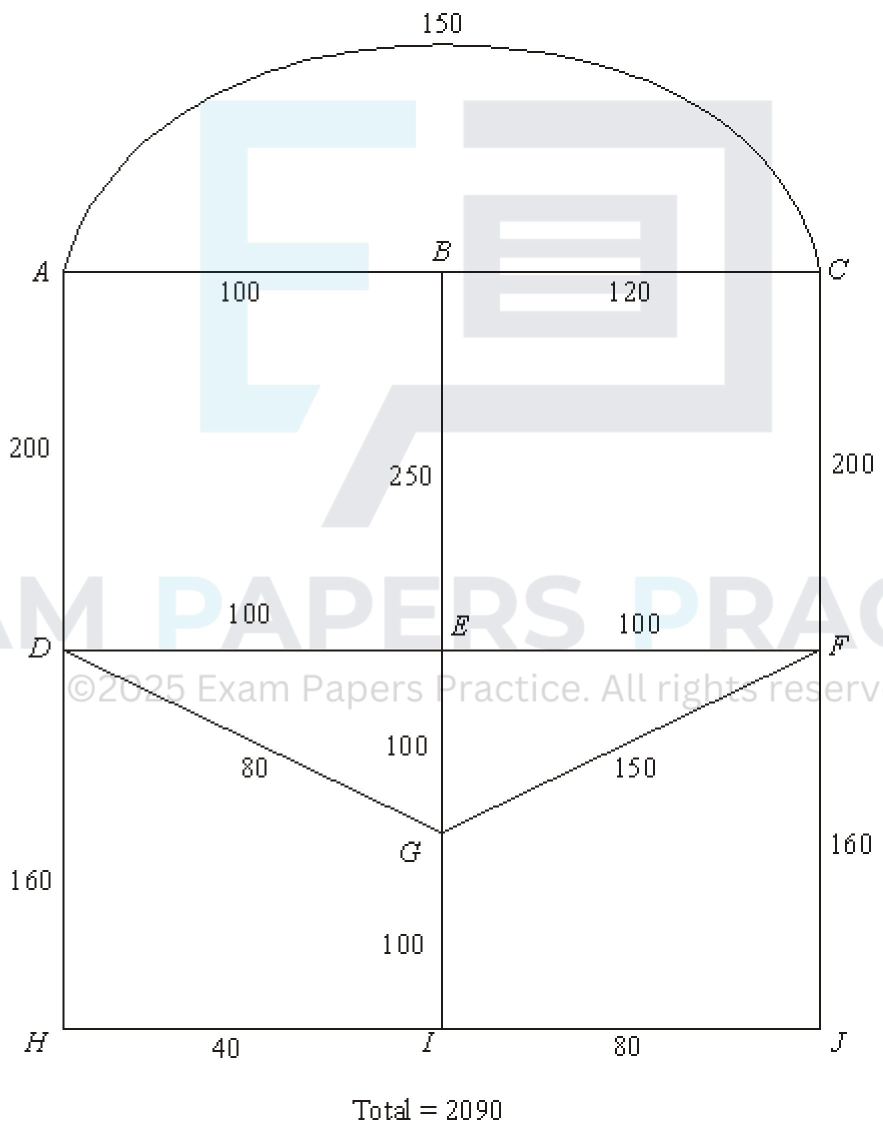
(ii) State the points from which Millie could start in order to achieve this optimal route.

(1)

(Total 11 marks)

Q12.

Stella is visiting Tijuana on a day trip. The diagram shows the lengths, in metres, of the roads near the bus station.



Stella leaves the bus station at A. She decides to walk along all of the roads at least once before returning to A.

(a) Explain why it is not possible to start from A, travel along each road only once and return to A.

(1)

(b) Find the length of an optimal 'Chinese postman' route around the network, starting and finishing at A.

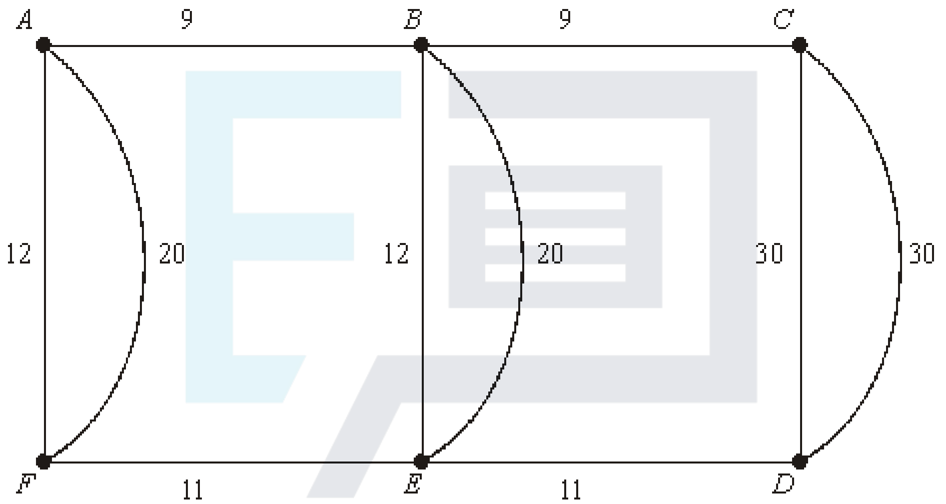
(5)

(c) At each of the 9 places B, C, ..., J, there is a statue. Find the number of times that Stella will pass a statue if she follows her optimal route.

(2)
(Total 8 marks)

Q13.

The diagram shows a network of roads connecting 6 villages. The number on each edge is the length, in miles, of the road.



Total length of the roads = 164 miles

(a) A police patrol car based at village A has to travel along each road at least once before returning to A. Find the length of an optimal 'Chinese postman' route for the police patrol car.

(6)

(b) A council worker starts from A and travels along each road at least once before finishing at C. Find the length of an optimal route for the council worker.

(2)

(c) A politician is to travel along all the roads at least once. He can start his journey at any village and can finish his journey at any village.

(i) Find the length of an optimal route for the politician.

(2)

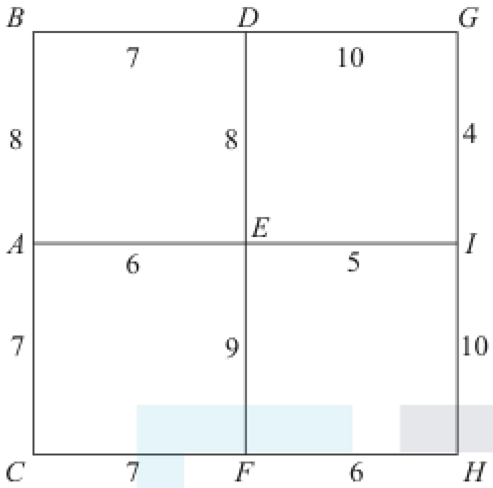
(ii) State the vertices from which the politician could start in order to achieve this optimal route.

(1)
(Total 11 marks)

Q14.

A local council is responsible for gritting roads.

The diagram shows the length, in miles, of the roads that have to be gritted.



Total length = 87 miles

The gritter is based at A, and must travel along all the roads, at least once, before returning to A.

(a) Explain why it is not possible to start from A and, by travelling along each road only once, return to A.

(1)

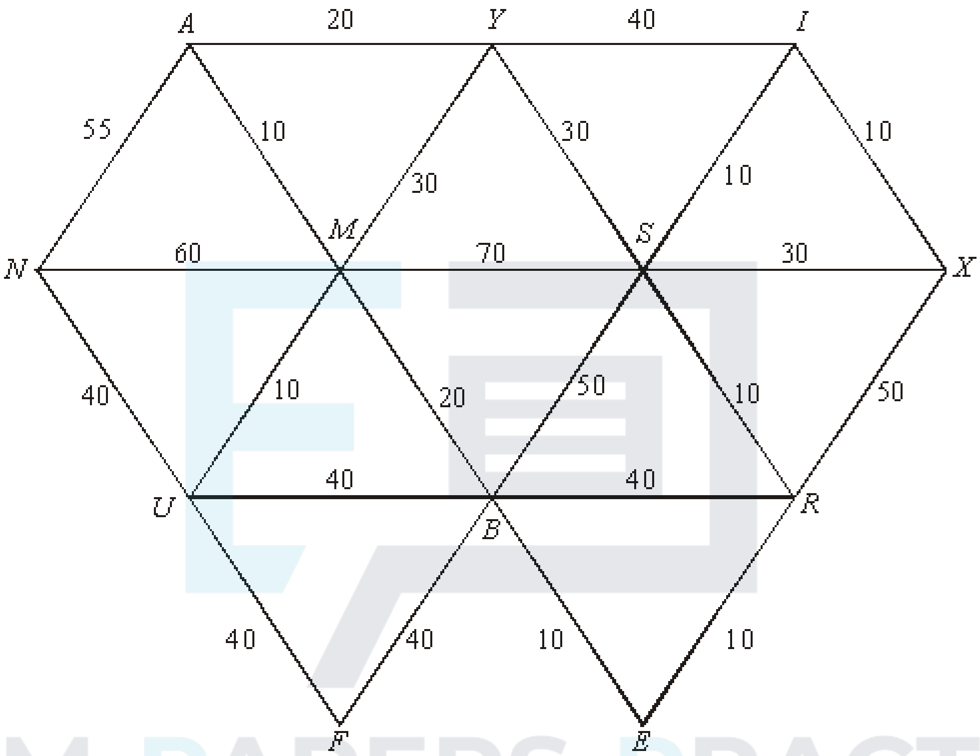
(b) Find an optimal 'Chinese postman' route around the network, starting and finishing at A. State the length of your route.

(6)

(Total 7 marks)

Q15.

Patrick has been sent as a prisoner to a new prison, which is in the form of a remote village. The diagram shows the houses where the prisoners live and paths connecting them. The number on each path represents the time, in seconds, it takes for Patrick to walk along that path. Patrick lives in house N.



On a day when all the paths are clear, Patrick walks along all the paths, looking for a possible escape route.

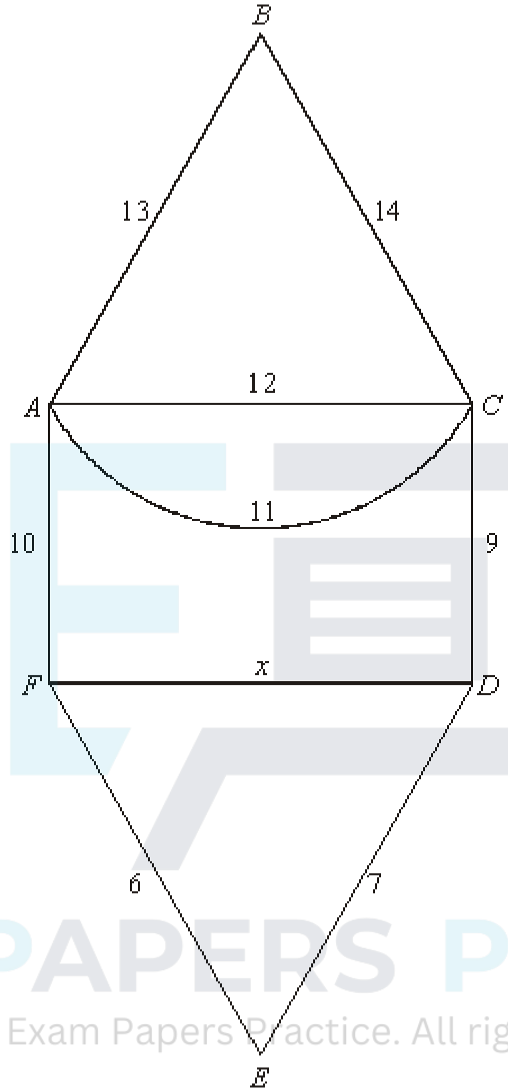
The total of all the times on the diagram is 725 seconds.

Find the minimum time taken for Patrick to walk along all the paths, starting and finishing at his house, N.

(Total 4 marks)

Q16.

The following diagram shows a network of roads connecting six towns. The number on each arc represents the distance, in miles, between towns. The road connecting towns D and F has length x miles, where $x < 13$.

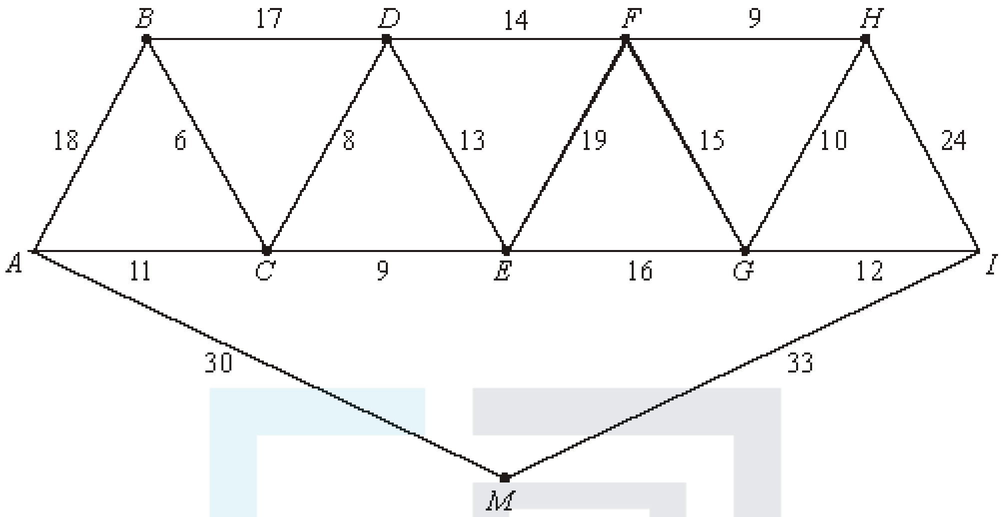


An optimal Chinese Postman route, starting and finishing at A, has length 100 miles. Find the value of x.

(Total 4 marks)

Q17.

Every Christmas Sam puts lights on nine trees, A, B, C, D, E, F, G, H and I, in his garden. The trees have to be connected together either directly or indirectly by cabling and also connected to the mains electricity supply, M, at the house. The cabling has to be laid alongside the garden paths. The diagram shows the lengths of the paths, in metres.



The total length of the paths is 264 metres.

(a) (i) Use Kruskal's algorithm to find the minimum length of cabling required. State this minimum length.

(5)

(ii) Draw the minimum spanning tree.

(2)

(b) The table shows the minimum lengths of paths between A, B, H and I.

	A	***B***	***H***	***I***
A	–	17	42	48
B	17	–	37	43
H	42	37	–	22
I	48	43	22	–

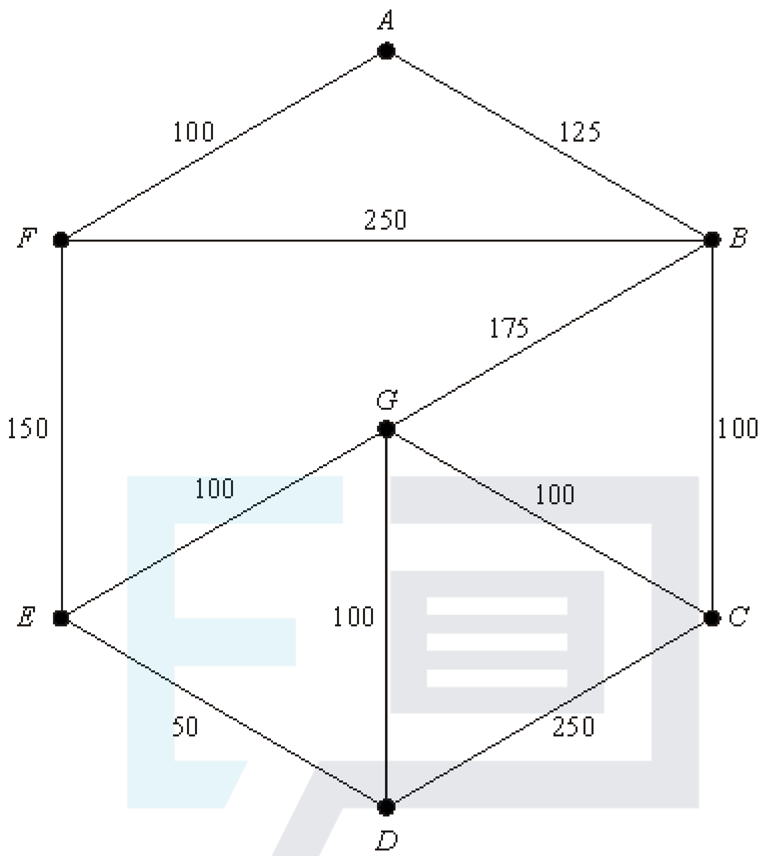
Sam decides to walk along all the paths, starting and finishing at M.

Find the length of an optimal Chinese Postman route for Sam.

(4)
(Total 11 marks)

Q18.

Brendan sells fish from his van. He travels around the streets of an estate. The diagram shows the network of streets he travels along. The number on each arc represents the length, in metres, of that street.



(a) (i) Give a reason why it is **not** possible to start at A, travel along each street once only and return to A.

(1)

(ii) Find the length of an optimal Chinese Postman route around the network, starting and finishing at A. State the number of times each vertex is visited.

(7)

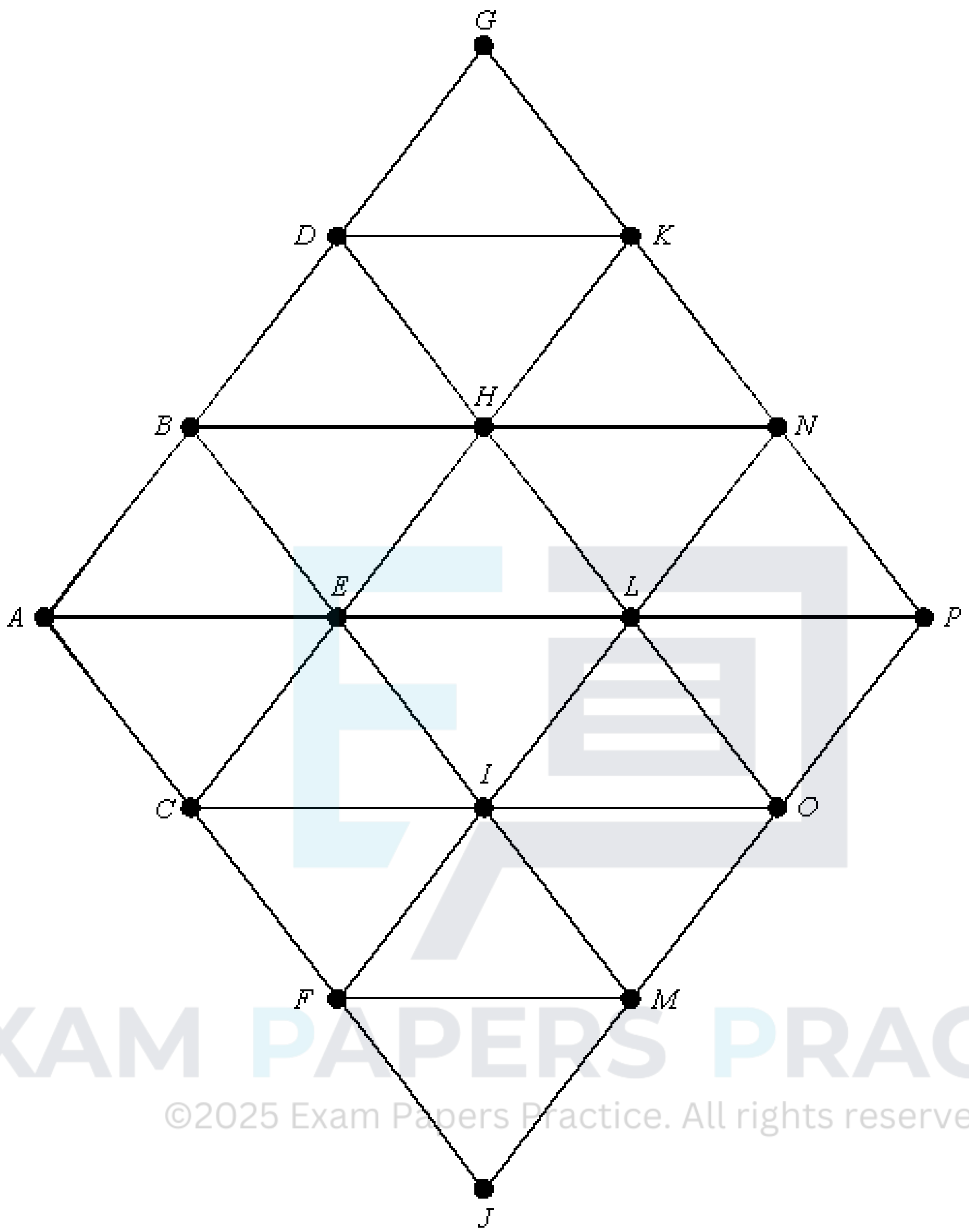
(b) The streets of another estate have 6 odd vertices. Find the number of ways of pairing these odd vertices.

(2)

(Total 10 marks)

Q19.

The following network has 16 vertices.



(a) Given that the length of each edge is 1 unit, find:

(i) the shortest distance from A to K;

(1)

(ii) the length of a minimum spanning tree.

(2)

(b) (i) Find the length of an optimal Chinese Postman route, starting and finishing at A.

(4)

(ii) For such a route, state the edges that would have to be used twice.

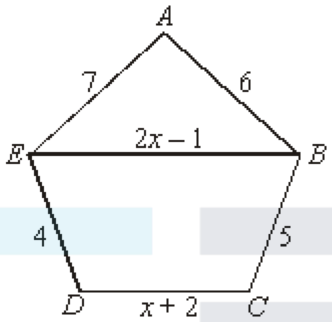
(1)

(iii) Given that the edges AE and LP are now removed, find the new length of an optimal Chinese Postman route, starting and finishing at A.

(2)
(Total 10 marks)

Q20.

The following diagram shows a network of roads connecting five villages. The numbers on the roads are the times, in minutes, taken to travel along each road, where $x > 0.5$.



A police patrol car has to travel from its base at B along each road at least once and return to base.

(a) Explain why a route from B to E must be repeated.

(1)

(b) List the routes, and their lengths, from B to E, in terms of x where appropriate.

(2)

(c) On a particular day, it is known that $x = 10$.

Find the length of an optimal Chinese Postman route on this day. State a possible route corresponding to this minimum length.

(4)

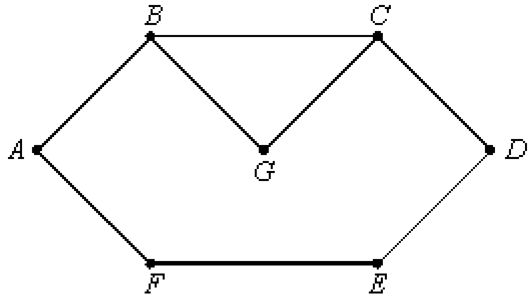
(d) Find, no matter what the value of x, which of the three routes should **not** be used if the total length of a Chinese Postman route is to be optimal.

(5)
(Total 12 marks)

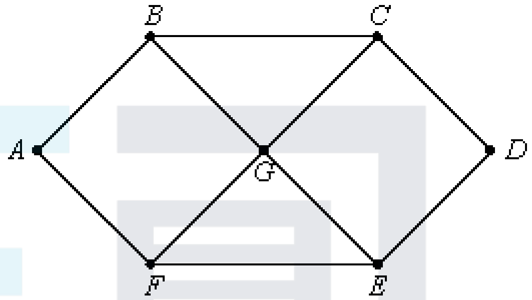
Q21.

The following question refers to the three graphs: **Graph 1**, **Graph 2** and **Graph 3**.

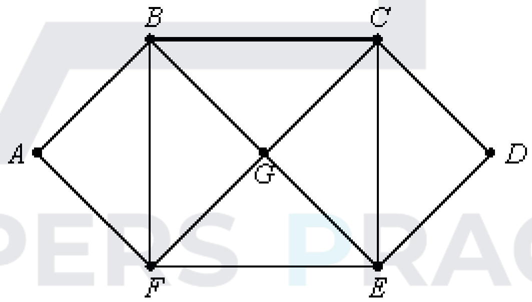
Graph 1



Graph 2



Graph 3



(a) For **each** of the graphs explain whether or not the graph is Eulerian.

(4)

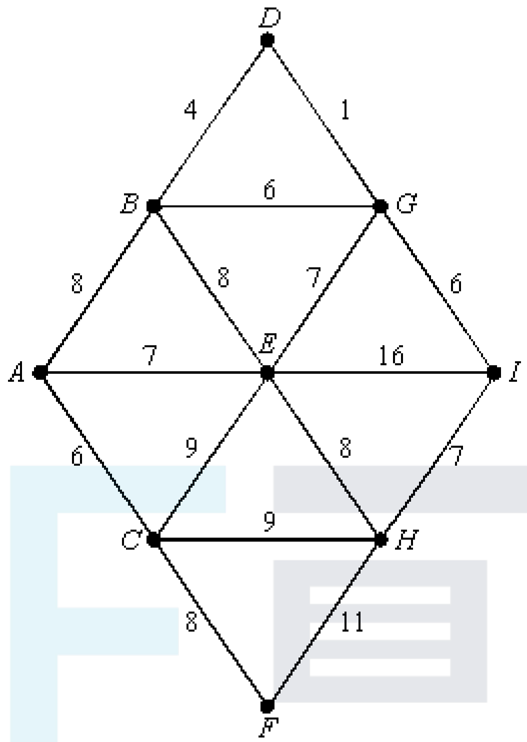
(b) The length of each edge connecting two vertices is 1 unit. Find, for **each** of the graphs, the length of an optimal Chinese postman route, starting and finishing at A.

(4)

(Total 8 marks)

Q22.

The following diagram shows the lengths of roads, in miles, connecting nine towns.



(a) Use Prim's algorithm starting from A, showing your working at each stage, to find the minimum spanning tree for the network. State its length.

(6)

(b) (i) Find an optimal Chinese Postman route around the network, starting and finishing at A. You may find the shortest distance between any two towns by inspection.

(4)

(ii) State the length of your route.

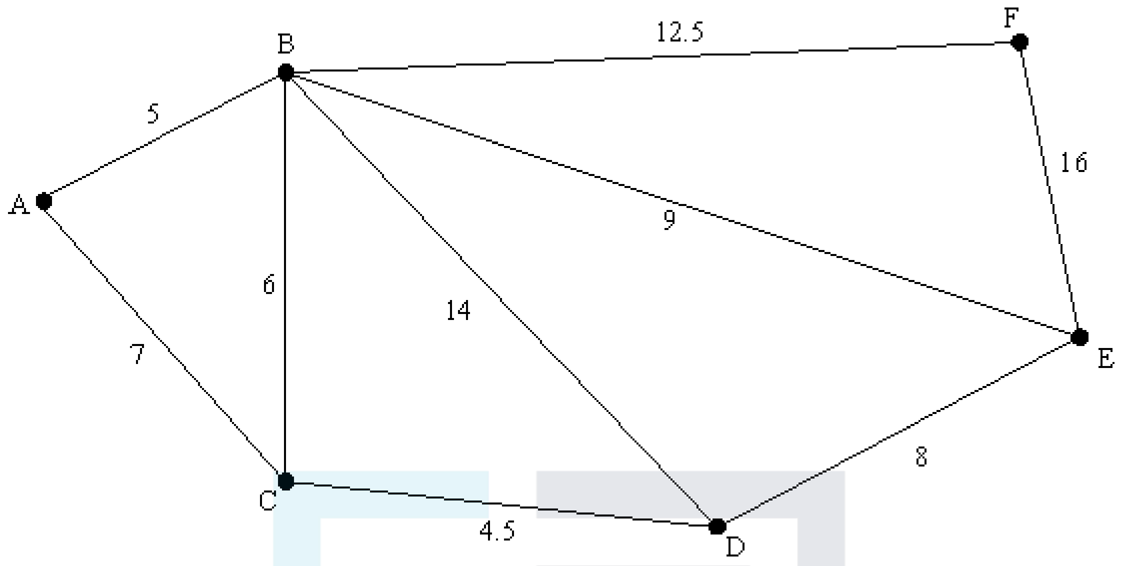
(1)

(Total 11 marks)

Q23.

A local council is responsible for gritting roads.

(a) The following diagram shows the lengths of roads, in miles, that have to be gritted.



The gritter is based at A and must travel along all the roads, at least once, before returning *to* A.

(i) Explain why it is **not** possible to start from A and, by travelling along each road only once, return to A.

(1)

(ii) Find an optimal 'Chinese postman' route around the network, starting and finishing at A. State the length of your route.

(6)

(b) (i) The connected graph of the roads in the area run by another council has six odd vertices. Find the number of ways of pairing these odd vertices.

(1)

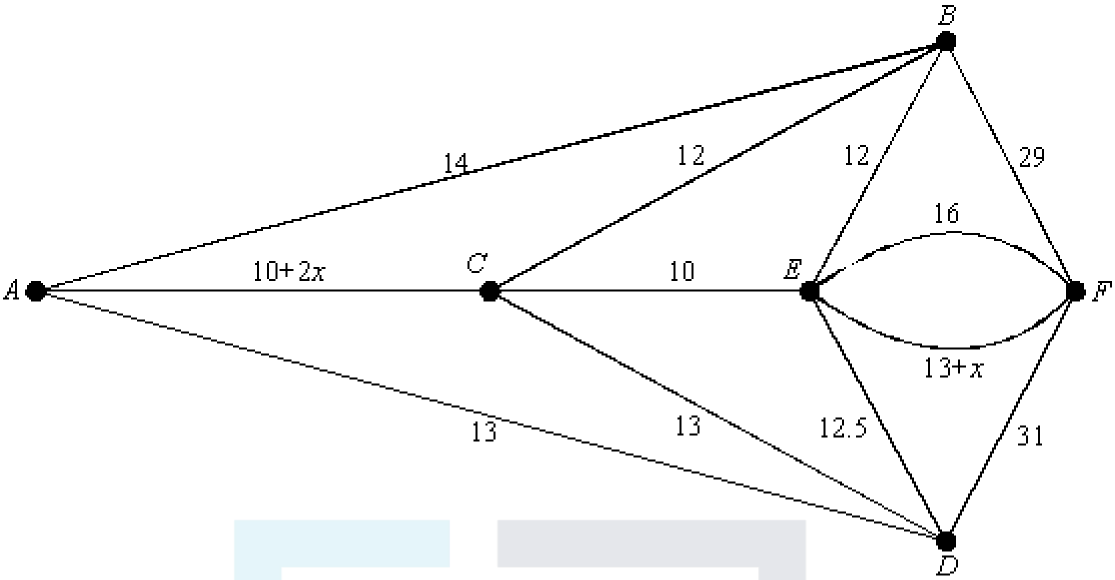
(ii) For a connected graph with n odd vertices, find an expression for the number of ways of pairing these vertices.

(2)

(Total 10 marks)

Q24.

The diagram shows the time, in minutes, for a traffic warden to walk along a network of roads, where $x > 0$.



The traffic warden is to start at A and walk along each road at least once, before returning to A.

(a) Explain why a section of roads from A to E has to be repeated.

(1)

(b) The route ACE is the second shortest route connecting A to E. Find the range of possible values of x.

(2)

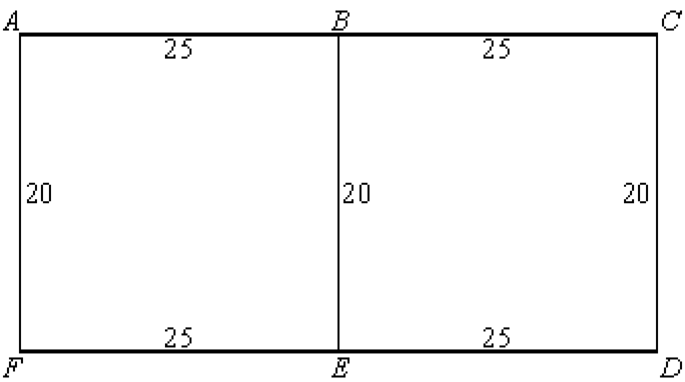
(c) Find, in terms of x, an expression for the minimum distance that the traffic warden must walk and write down a possible route that he could take.

(3)

(Total 6 marks)

Q25.

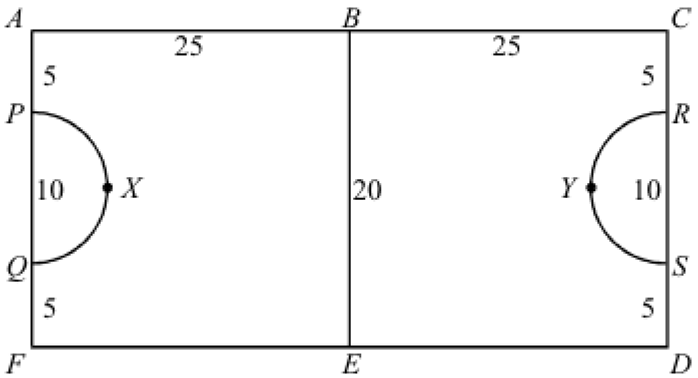
A groundsman at a local sports centre has to mark out the lines of several five-a-side pitches using white paint. He is unsure as to the size of the goal area and he decides to paint the outline as given below, where all the distances given are in metres.



(a) He starts and finishes at the point A. Find the minimum total distance that he must walk and give one of the corresponding possible routes.

(3)

(b) Before he starts to paint the second pitch he is told that each goal area is a semi-circle of radius 5 metres, as shown in the diagram below.



(i) He can start at any point but must return to his starting point. State which vertices would be suitable starting points to keep the total distance walked from when he starts to paint the lines until he completes this task to a minimum.

(2)

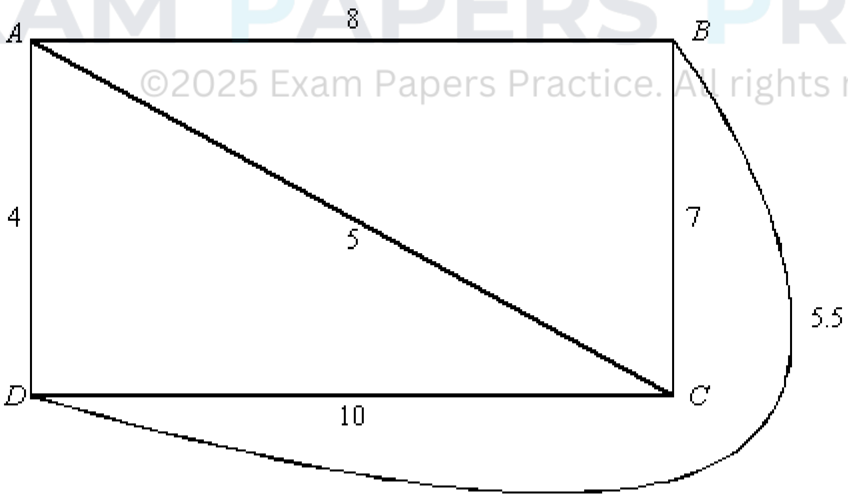
(ii) Find an optimal 'Chinese postman' route around the lines. Calculate the length of your route.

(3)

(Total 8 marks)

Q26.

In the following network the four vertices are odd.



(a) List the different ways in which the vertices can be arranged as two pairs.

(2)

(b) For **each** pairing you have listed in part (a), write down the sum of the shortest distance between the first pair and the shortest distance between the second pair. Hence find the length of an optimal 'Chinese postman' route around the network.

(3)

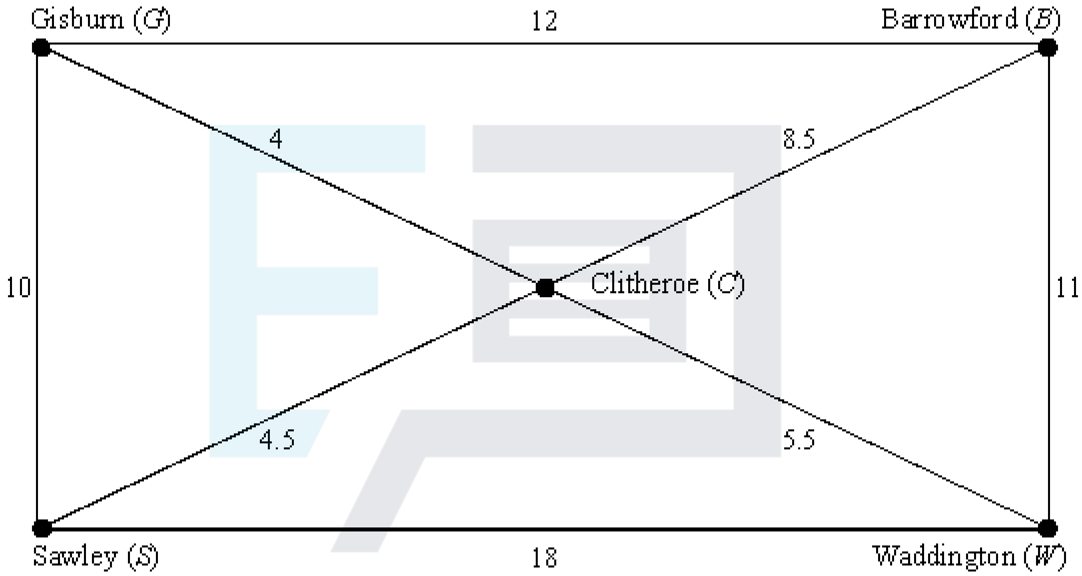
(c) State the minimum number of extra edges that would need to be added to the network to make the network Eulerian.

(1)
(Total 6 marks)

Q27.

A highways department has to inspect its roads for fallen trees.

(a) The following diagram shows the lengths of the roads , in miles, that have to be inspected in one district.



(i) List the three different ways in which the four odd vertices in the diagram can be paired

(1)

(ii) Find the shortest distance that has to be travelled in inspecting all the roads in the district, starting and finishing at the same point.

(4)

(b) The connected graph of the roads in another district has six odd vertices. Find the number of ways of pairing these odd vertices.

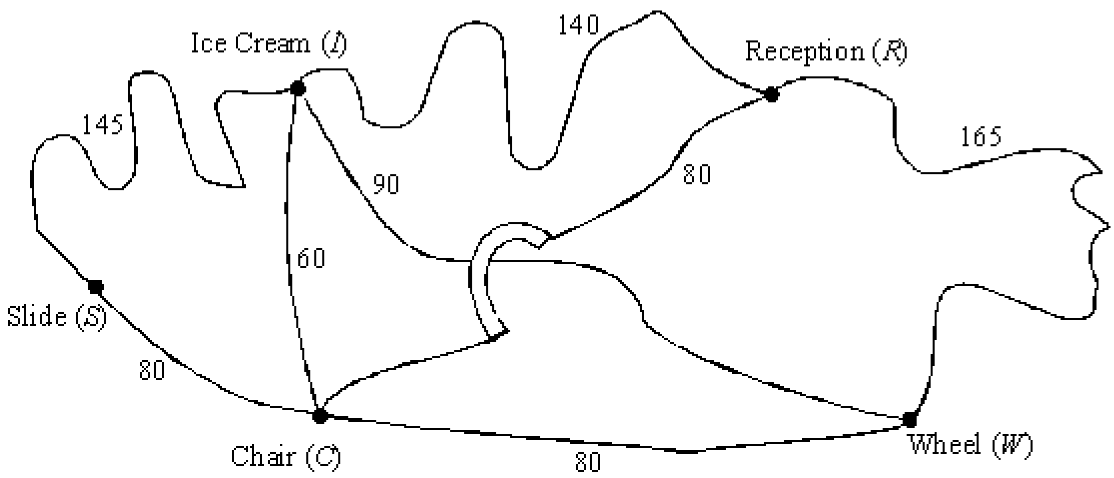
(2)

(c) For a connected graph with n odd vertices, find an expression for the number of ways of pairing these odd vertices.

(3)
(Total 10 marks)

Q28.

A theme park employs a student to patrol the paths and collect litter. The paths that she has to patrol are shown in the following diagram, where all distances are in metres. The path connecting I and W passes under the bridge which carries the path connecting C and R.



(a) (i) Find an optimal "Chinese postman" route that the student should take if she is to start and finish at Reception (R).

(2)

(ii) State the length of your route.

(1)

(b) (i) A service path is to be constructed. Write down the two places that this path should connect, if the student is to be able to walk along every path without having to walk along any path more than once.

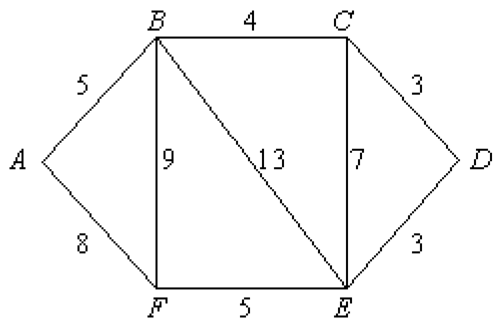
(2)

(ii) The distance walked by the student in part (b)(i) is shorter than that found in part (a)(ii). Given that the length of the service path is l metres, where l is an integer, find the greatest possible value of l.

(2)

(Total 7 marks)

Q29.



(a) The network represents a road system in which the lengths of the roads are shown in kilometres. The road system has to be cleared of snow by a snow plough which is based at A. The snow plough only needs to travel along each road once in order to clear it. However, when all roads have been cleared it must return to its base at A.

(i) State which roads the snow plough should drive along twice in order to travel the minimum total distance.

(2)

(ii) Hence solve the Chinese postman problem for this network to find a route for the snow plough, starting and finishing at A.

(2)

(iii) The road BE becomes a dual carriageway and must be cleared twice, once in each direction. Redraw the network to take account of this and find, by inspection, a new route for the snow plough.

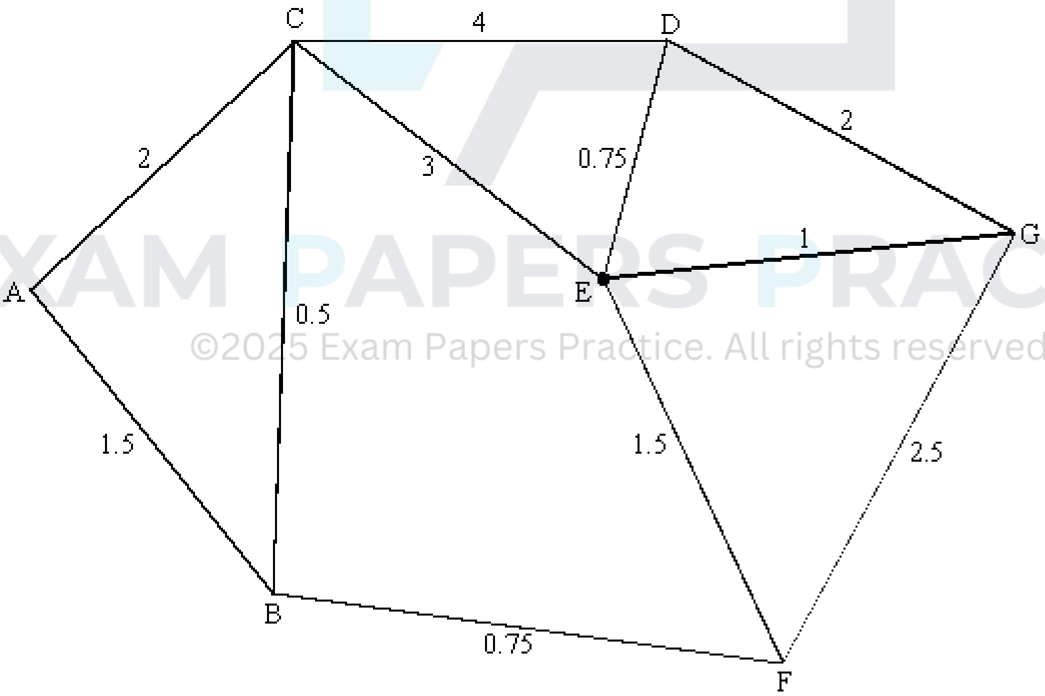
(3)

(b) The original network has to be drawn as efficiently as possible by a pen operated by a computer. Explain how this can be done without lifting the pen from the paper and without tracing any of the edges more than once.

(2)
(Total 9 marks)

Q30.

A road gritting service is based at a point A. It is responsible for gritting the network of roads shown in the diagram, where the distances shown are in miles.



(a) Explain why it is **not** possible to start from A and, by travelling along each road only once, return to A.

(2)

(b) In the network there are four odd vertices, B, D, F and G. List the different ways in which these odd vertices can be arranged as two pairs.

(2)

(c) For **each** pairing you have listed in part (b), write down the sum of the shortest distance between the first pair and the shortest distance between the second pair.

(3)

(d) Hence find an optimal "Chinese postman" route around the network, starting and finishing at A. State the length of your route.

(3)

(Total 10 marks)

Q31.

The towns P, Q, R, S, T, U, V and W are all linked to each other by direct roads whose lengths, in miles, are shown in the following table:

	P	Q	R	S	T	U	V	W
P	–	5	4	3	4	5	4	6
Q	5	–	7	5	3	5	8	8
R	4	7	–	6	8	6	3	5
S	3	5	6	–	4	6	9	6
T	4	3	8	4	–	7	8	6
U	5	5	6	6	7	–	8	3
V	4	8	3	9	8	8	–	8
W	6	8	5	6	6	3	8	–

(a) An estate agent wants to start at the town P and visit each of the other towns once before returning to P.

(i) Use the nearest neighbour algorithm to find one possible route for the estate agent and state its total length.

(4)

(ii) On the table below, ignore the town P and find the length of a minimum connector of the towns Q, R, S, T, U, V and W.

	P	Q	R	S	T	U	V	W
P	–	5	4	3	4	5	4	6
Q	5	–	7	5	3	5	8	8
R	4	7	–	6	8	6	3	5
S	3	5	6	–	4	6	9	6
T	4	3	8	4	–	7	8	6

U	5	5	6	6	7	–	8	3
V	4	8	3	9	8	8	–	8
W	6	8	5	6	6	3	8	–

(4)

(iii) Deduce a lower bound for the estate agent's route and comment on its significance to your answer to part (a)(i).

(3)

(b) The estate agent would then like to look at the houses on each of the roads joining the towns. The estate agent starts at P, covers each of the roads at least once, and finishes at P.

(i) Explain why it will be necessary to cover at least four of the roads more than once.

(2)

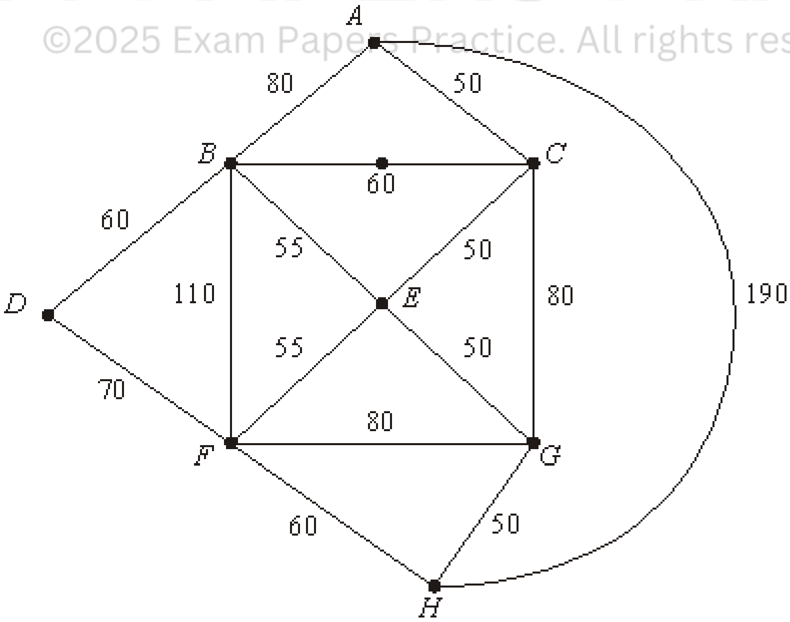
(ii) Given that the total length of all the roads is 160 miles, explain how you know that all the roads can be covered with a round trip of 172 miles.

(3)

(Total 16 marks)

Q32.

Eight alpine huts $A - H$ are connected by footpaths, as shown in the following network. The numbers denote the lengths of the paths, in metres.



The total length of all the footpaths is 1100 metres.

(a) A snowplough, starting at A, needs to travel along each of the footpaths at least once, and finish back at A.

(i) Explain how you know that it will be necessary for the snowplough to go along some footpaths more than once.

(1)

(ii) Apply the Chinese postperson algorithm to find the shortest distance which the snowplough will have to travel. (You may find the shortest distances between pairs of vertices by inspection.)

(4)

(iii) Give an example of a suitable shortest route.

(3)

(b) The caretaker of the huts wants to start at A, pass each hut exactly once, and finish back at A.

(i) Use a nearest neighbour approach to find a suitable route of length 470 metres.

(3)

(ii) Explain why, if the caretaker uses footpath AH, the total length of his route will be more than (190 + (7 × 50)) metres.

(1)

(iii) Show that, if the caretaker does not use footpath AH, his route must include footpaths AB, AC, DB, DF, HF and HG. Hence show that the caretaker's route which you found in part (b)(i), or its reverse, is the caretaker's only route which does not use footpath AH.

(2)

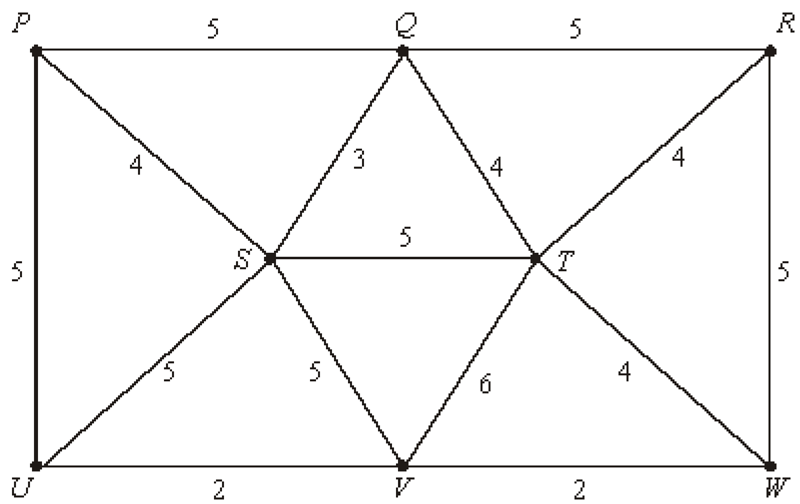
(iv) Deduce that the caretaker's route which you found in part (b)(i) is the shortest possible.

(2)

(Total 16 marks)

Q33.

The network shows eight Chinese railway stations P – W and the lengths, in miles, of the tracks linking them:



(a) The Chinese Trackrail inspector wishes to travel by rail and check the whole network by starting at P, travelling along each track at least once, and ending back at P.

(i) Explain why it will be necessary to repeat at least three tracks.

(2)

(ii) The inspector wishes to minimise the distance he has to travel. Determine which tracks he must repeat, given that they will include PS or PU.

(5)

(b) Find the length of the minimum connector of the network and illustrate this minimum connector as a tree.

(4)

(c) A train-spotter has to travel exactly 25 miles to get from his home to each of the stations P–W. He wishes to travel from home to a station, then by train to another station and another and so on until he has visited all the stations once. Then he will return home.

(i) Use your answer to part (b) to show that his round trip must be at least 73 miles in length.

(2)

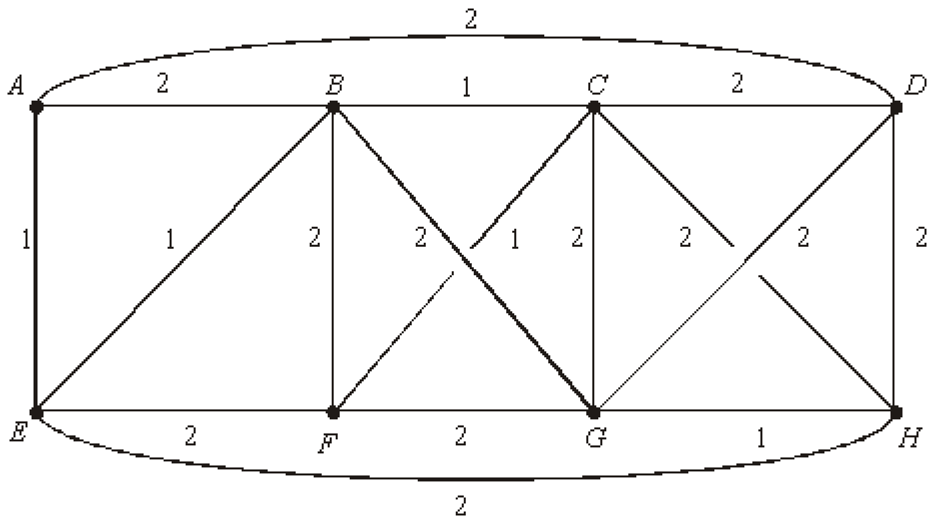
(ii) Explain why a route of 73 miles is not possible and, by adapting your minimum connector, find a round trip of length74 miles for the train-spotter.

(3)

(Total 16 marks)

Q34.

The network below shows the phone lines connecting eight offices A–H. The cost per minute for a phone call between two offices is either 1p or 2p as shown on the network.



(a) For the transfer of information, the offices wish to keep a cycle of eight phone lines open. In this way one office will be linked to another, which will be linked to another, and so on until the eighth one is then linked back to the first.

(i) Use a nearest neighbour approach, starting at A, to find one suitable cycle of phone lines to keep open.

(3)

(ii) Explain why the cycle which you found in part (i) is in fact the cheapest possible.

(1)

(b) The phone company wishes to check all the lines shown in the above network by starting at A, passing a message along one line, and then the receiver of the message will pass it along another line, and so on until the message is eventually received back at A, having used each of the lines at least once.

(i) Use the Chinese postperson algorithm to find which lines will be used twice when this checking is done in the most economical way.

(4)

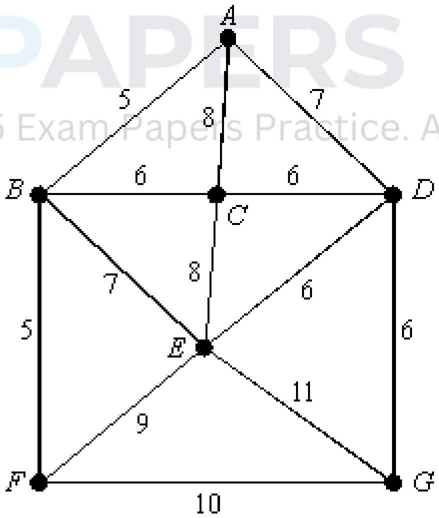
(ii) The condition that the message ends back at A is dropped, and the company merely requires a route which uses each line at least once. In the new most economical route, which line, or lines, will be used more than once, and at which office will the message end?

(2)

(Total 10 marks)

Q35.

The numbers of parking meters on the roads in a town centre are shown in the following network.



(a) A traffic warden wants to start at A, walk along the roads passing each meter at least once, and finish back at A. She wishes to choose her route in order to minimise the number of meters that she passes more than once.

(i) Explain how you know that it will be necessary for her to pass some meters more than once.

(1)

(ii) Apply the Chinese postperson algorithm to find the minimum number of meters which she will have to pass more than once, and give an example of a suitable route.

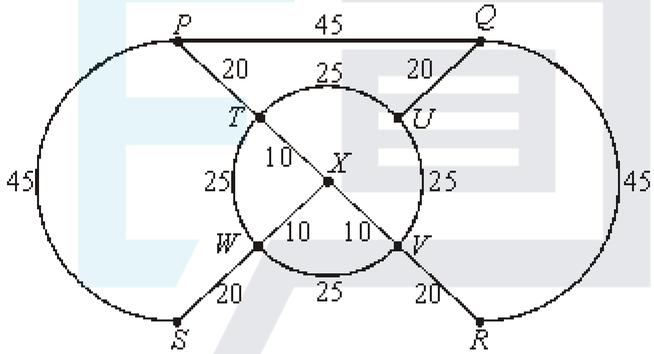
(6)

(b) At each of the junctions A, B, C, D, E, F and G there is a set of traffic lights. The traffic warden is asked to make a journey, starting and finishing at A, to check that each set of traffic lights is working correctly. Find a suitable route for her which passes 50 or fewer meters.

(4)
(Total 11 marks)

Q36.

(a) The vertices of the following network represent the chalets in a small holiday park and the arcs represent the paths between them, with the lengths of the paths given in metres.



A gardener wishes to sweep all the paths, starting and finishing at P, and to do so by walking (always on the paths) as short a distance as possible. Apply the Chinese postperson algorithm to find the shortest distance the gardener must walk, and give one possible shortest route.

(5)

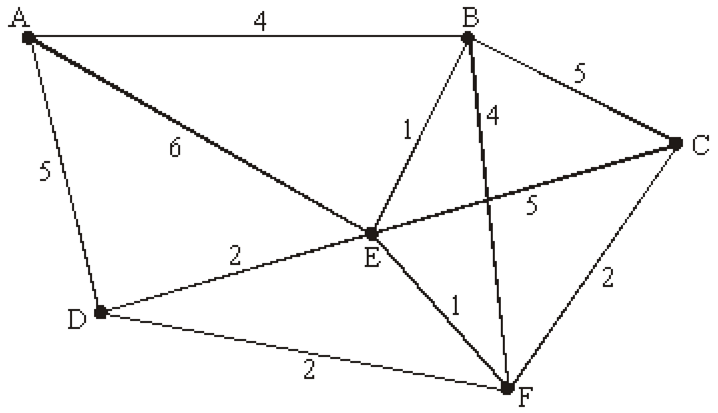
(b) In the holiday park described in part (a) the path from W to V is closed and is replaced by a new path from S to R. Consider the corresponding network obtained after these changes. State the number of odd vertices in the new network and state how many pairings of these vertices would need to be considered in applying the Chinese postperson algorithm.

(Do **not** actually apply the algorithm.)

(2)
(Total 7 marks)

Q37.

The network below has distances marked in its edges.



(a) Use an appropriate algorithm (not complete enumeration) to find the shortest route and distance from A to C. Give your route and its length.

Show your working and briefly describe the steps of your algorithm.

(7)

(b) Use your working from part (a) to write down the shortest routes and distances from A to each of the remaining vertices.

(3)

(c) Find a minimum length path traversing each edge of the network at least once, and returning to the start vertex A. (You may find the shortest distances other than from vertex A by inspection.)

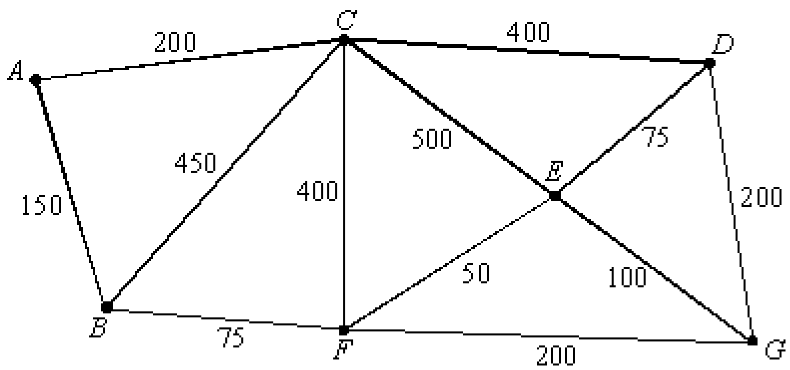
Show your working and briefly describe the steps of your algorithm.
Give your route and its length.

(4)

(Total 14 marks)

Q38.

The graph is a representation of a system of roads. The lengths of the roads are shown in metres.



(a) Use a simple algorithm (not complete enumeration) to find the shortest route from A to G, and give the length of that shortest route.

You must show sufficient detail to demonstrate your use of the algorithm. The answer alone will not gain credit.

(5)

(b) Explain why the graph is not Eulerian.

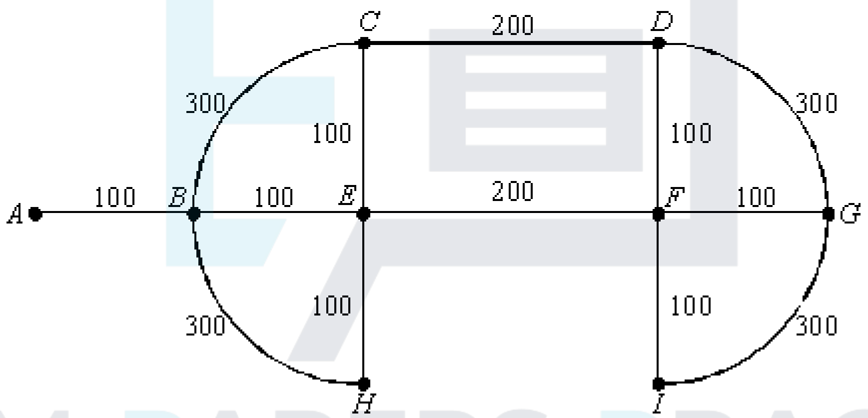
By considering ways of pairing the odd vertices, find a shortest route, starting and finishing at A, which traverses each road at least once. State the length of your route.

(6)
(Total 11 marks)

Q39.

The edges of the network below represent roads in a small housing estate. There is only one road into and out of the estate, represented by edge AB. The lengths of the roads are shown in metres.

The total length of roads is 2300 m.



(a) A papergirl delivers to about 20% of the houses and therefore finds it worthwhile to cross from side to side of a road whilst making deliveries. Thus, she would **prefer** to walk along each road only once.

Use an appropriate algorithm to find the minimum total length of road along which she must walk if she starts and finishes her round at A. Indicate how you applied the algorithm.

(4)

(b) A postman delivers to about 80% of the houses. He finds it better to deliver to one side of a road at a time. He therefore needs to walk along each road at least twice. If he starts and finishes at A, find the minimum distance that he must walk, justifying your answer.

(3)

(c) Suppose that the postman acquires a bicycle. This means that he would still like to travel along each road twice, but in opposite directions, so that he is always riding on the correct side of the road. Will this mean that he must travel further? Justify your answer.

(2)
(Total 9 marks)